



Caught on Camera: Toward Evaluating and Defending On-screen Deepfakes With Mobile and Wearable Camera Devices

Shuhao Zhang¹, Pai Zheng¹, Yuanzhe Yang², Qinhong Jiang³, and Yan Long^{1*}

¹ The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China

² National University of Singapore, Singapore

³ The Hong Kong Polytechnic University, Hong Kong SAR, China

{szhang515, pzheng585}@connect.hkust-gz.edu.cn

e1397069@u.nus.edu; qinhong.jiang@polyu.edu.hk

yanlong@hkust-gz.edu.cn

Abstract. Existing deepfake detection is assumed to operate solely in the *digital regime* where the detector has direct access to generated fake media files. This paper investigates an emerging new threat model in the mobile and wearable computing era, where users encounter and verify suspicious content displayed on adversary-controlled screens through smartphone and smart glasses cameras. Failure or unreliable detection of on-screen in public settings could amplify misinformation and mislead high-stakes decision-making. Given this gap of *display-recapture* deepfake detection model, which has received limited systematic investigation, we present a large-scale measurement of display-mediated deepfake attacks across 103 recaptured experimental configurations, spanning two smartphones, two smart-glasses devices, six public display devices, and variations in viewing distance, angle, lighting, and other impactful imaging conditions. Tests with three most representative deepfake detectors on the Celeb-DF-v2 subset reveal significant detection performance degradation, sometimes down to random guesses. Based on controlled factor analysis and empirical modeling, we develop a Caught-on-Camera mitigation pipeline as a plug-and-play, selective data augmentation module that is capable of enhancing any existing detectors without collecting display and camera-specific data. Evaluation on 29,900 physically collected images shows an average accuracy increase from 65.6% to 89.5% on even the least robust detector. We further highlight the remaining gaps, such as unique time-varying color patterns of conference room projector images that lend adversaries extra advantages and require follow-up dedicated defensive research. In summary, we intend for this work to serve as a wake-up call for defensive attention to display-mediated deepfake threats in public-communication settings.

Keywords: deepfake detection · smart glasses · smartphones · display-recapture · physical-world presentation attacks · multimedia security

* Corresponding author.

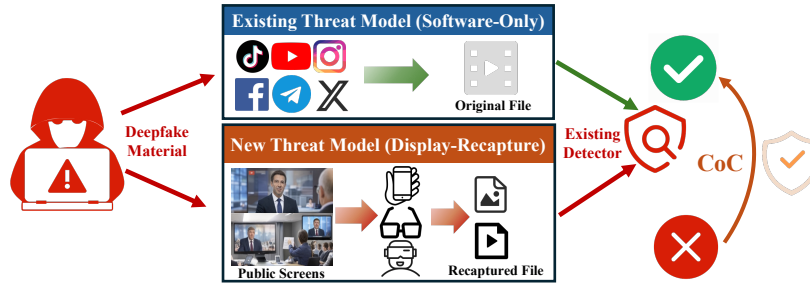


Fig. 1. Comparison between the conventional and emerging display-recapture deepfake detection threat models.

1 Introduction

Deepfakes refer to images and other multimedia content that have been edited or created by AI techniques, particularly with deep neural network-based generation models. Since the non-consensual face-swapped pornography images attracted wide public attention in 2017 [44], the emergence of low-cost, easily deployable deepfake generation pipelines has caused a broad array of significant security and social impacts, ranging from impersonation-based fraud to political misinformation, and a growing loss of trust in online images and videos. As the generation pipeline becomes open-source and widely available, deepfakes are no longer limited to online social platforms, but have become a growing concern for public communication, journalism, and legal evidence [35,30,39].

Deepfake detection capabilities have thus become an urgent and necessary need in modern multimedia systems. However, state-of-the-art detectors are built upon a *direct-recognition* threat model: given an image or video file generated by deepfake software, a detector classifies whether the content is real or deepfake-synthesized directly from the bit-stream [21]. Benchmarks such as FaceForensics++ [29] and DeepfakeBench [41] report accuracy and area under the receiver operating characteristic curve (AUC) on curated digital datasets, implicitly and unanimously assuming that test inputs are the same kind of bit-streams that generative models produce. This existing threat model is mostly oriented to large media platforms—such as content-sharing services (e.g., TikTok and X) that are expected to vet uploads at ingestion time—where *the original generated deepfake file is assumed to be directly available to the detector in the digital domain*.

In practice, however, a large and growing class of detection scenarios operates under fundamentally different conditions. Users encounter visual content via televisions, classroom projectors, public signage, and conference displays. When they become suspicious or intend to check the authenticity of the content, they must record the screen with a smartphone or, increasingly, with an augmented-reality (AR) device such as camera-equipped smart glasses [45,11], and only then invoke a detection tool on the user side. This creates a paradigm shift from *corporate platform-based detection to personal device-based deepfake detection*,

effectively moving the power of verification from large centralized platforms to end users. In these display-recapture workflows, the detector operates not on the original file but on a recaptured version shaped by physical imaging factors such as specular reflections, Moiré patterns, exposure shifts, geometric distortion, and camera-side image processing and re-compression. The content presented to the detector therefore has fundamentally different characteristics.

This shift also creates a practical incentive for attackers. If a detector can inspect the original generated file directly at upload time, the attacker must evade the detector in its native digital setting. By contrast, once verification occurs only after the content has been rendered on a physical display and recaptured by a user device, the attacker can rely on the display-recapture channel itself to degrade detector reliability. In other words, the attack surface is attractive not because it requires sophisticated new attack machinery, but because it turns an ordinary presentation path into an implicit evasion layer. As discussed later in Section 3.2, this threat is realistic in settings where suspicious content is observed first and verified only after display and recapture. We thus hypothesize that existing detectors, calibrated entirely on digital content and never designed to handle this distribution shift, would face critical performance degradation in this emerging display-recapture threat model.

Toward understanding the limits of existing detection algorithms and enhancing detector capabilities, our work formulates and characterizes *display-mediated deepfake attack*: an adversary who controls a display simply broadcasts off-the-shelf deepfake content and relies on the display-recapture channel to push recaptured inputs into the detector’s blind spots, without designing new generative models, adversarial perturbations, or hardware exploits. Our systematic analysis seeks to answer two interdependent research questions: **RQ1.** *To what degree do the physical factors in the display-recapture process degrade the performance of existing deepfake detectors, and which factors dominate?*; and **RQ2.** *What possible strategies may defenders use to mitigate the physical impacts and improve detection performance under the new threat model?*

Answering these questions poses an outstanding challenge. A naive approach of collecting recaptured data for every display-recapture combination to measure diverse factors’ impacts, and then retrain the detector models, is both impractical and technically infeasible without prior knowledge of the deployment environment. To address the challenge, our strategy instead identifies the dominant physical distortion factors through controlled single-factor experiments, designs augmentation operations that simulate each factor without collecting any physical data, and validates their combined benefit through a systematic three-stage pipeline. This approach makes the mitigation deployable without any device-specific data collection.

To verify the effectiveness of our defensive method in diverse real-world settings, we evaluate 103 recaptured configurations, including 19 controlled single-factor settings and 84 multi-factor settings, collected with 6 display devices and 4 camera devices comprising two smartphones and two smart-glasses cameras, with a total of 29,900 images collected for systematic evaluation. Guided by our

observations and modeling of how the display-recapture process distorts image data distribution, we provide a three-stage recapture-aware selective augmentation framework, named Caught-on-Camera (CoC), that can be applied to any existing deepfake detectors as a plug-and-play enhancement module. Our systematic evaluations show CoC yields substantial and consistent gains, improving Xception, one of the least robust detectors in the new display-recapture threat model, from 0.656 to 0.895 average frame-level accuracy on recaptured data, raising average AUC from 0.890 to 0.965, and improving performance across all 103 configurations.

We further analyze the remaining gaps and promising directions for future defensive research aiming to protect this emerging threat model, and showcase more challenging and unique display-recapture distortion channels of DLP projector contents that exhibit strong time-varying color banding distortions, which provide potential adversaries with even stronger instrumental leverage and call for immediate dedicated investigations. We summarize our key contributions of this work as follows:

1. **Threat model and uncovered performance gap.** We propose an increasingly important display-recapture threat model that uses user cameras to detect on-screen deepfakes in public spaces. The significant performance drop of prevalent deepfake detectors in this setting reveals a previously overlooked real-world security gap and serves as a wake-up call for immediate defensive attention for public good.
2. **Dedicated dataset¹ and factor characterization.** We construct a 103-configuration, 29,900-image dataset spanning two smartphones, two smart glasses devices, six display types, and systematic variation of viewing angle, distance, lighting, and Moiré conditions. To our knowledge, this is the most comprehensive physical-setting deepfake detection benchmark to date.
3. **Exemplary mitigation designs and strategy analysis.** We explore the device-agnostic design space for mitigating the physical recapture process’s impact without collecting additional training data. We also provide a detailed analysis of future works to further enhance the defensive designs.

2 Background and Motivation

2.1 Deepfake Image and Face-Swapping

Deepfakes were stimulated by the rapid progress of deep generative modeling after the introduction of generative adversarial networks (GANs) in 2014 [9]. Public awareness accelerated in late 2017, when the Reddit account “deepfakes” circulated celebrity face-swapped pornography and helped turn deepfake from the name of a specific face-swap workflow into a broader label for AI-generated fake

¹ ScreenDeepfakeBench repository:
<https://github.com/SENSE-Lab-Security/ScreenDeepfakeBench>.

media [5]. Today, the technical space of deepfake images and videos includes several categories, such as entire-face synthesis, attribute manipulation, and expression manipulation [44,34,20]. Among these categories, our paper focuses on face swapping, which remains one of the most prominent and widespread identity-editing techniques in major facial forgery benchmarks [47,41,29,12]. Face swapping is also the subfield through which deepfakes first became socially salient, including fake videos of Tom Cruise [23] and Volodymyr Zelenskyy [36] that went viral and caused significant public confusion.

Deepfake face-swapping generation pipelines often consist of building blocks of face detection and alignment, feature replacement, targeted area blending, and post-processing steps such as color correction and blurring [14]. This pipeline is also broadly representative of other deepfake image techniques [34], where the content of replacement is changed (e.g., expression instead of identity), but still relies on those reusable building blocks. Therefore, the security gaps and mitigations this paper investigates on face-swapping deepfakes could be transferable and generalized to other image and video deepfakes.

2.2 Deepfake Detection

Deepfake detection is typically framed as a binary classification problem, where a model trained on genuine and manipulated content learns to distinguish the two classes using spatial, frequency, or temporal cues. Representative detectors that are currently mostly used in benchmarks and real-world deployment include Xception [29], which applies depthwise separable convolutions on aligned face crops, FFD [6], which incorporates face-level attention, and SPSL [13], which operates in the spatial-phase frequency domain and has shown strong cross-dataset generalization [41]. Performance is often evaluated on widely accepted benchmarks such as FaceForensics++ [29] and Celeb-DF [12]. Under these settings, existing detectors often report very strong performance, and in some in-domain evaluations, their accuracy can approach saturation, creating the impression that deepfake detection is largely a solved problem in the digital domain.

Nevertheless, we highlight that previous deepfake detection datasets and evaluation pipelines are fundamentally built upon the assumption of direct access to the generated deepfake file in the digital domain. This assumption aligns well with platform-level defenders, such as social media and content-sharing services, but differs substantially from the emerging display-recapture threat model, in which a user encounters suspicious content on a screen and must verify it by recording the screen with a smartphone or smart-glasses camera. Our work thus aims to address this clear gap in datasets and mitigation designs.

3 Problem Formulation

This section formulates the new threat model of display-mediated deepfakes and camera recapture-based detection.

3.1 Threat Model

The threat model consists of an attacker $\mathcal{A}$ and a defender $\mathcal{D}$.

Attacker. We assume $\mathcal{A}$ controls one or more public displays and schedules arbitrary visual content on them, using off-the-shelf deepfake generation tools. The attacker does not have any knowledge about the defender’s detection tools; the sole attack action is broadcasting existing deepfake content through a physical display channel that detectors were never trained to handle. While more advanced adversarial example-style attacks that generate delicate perturbations to bypass detection are also feasible under the display-mediated threat model, this work mostly considers low-cost, low-effort *zero-optimization* attacks that any average computer user can launch with access to existing open-source or commercial deepfake generation software.

Defender. We assume $\mathcal{D}$ is any ordinary bystander who observes the display and records it with personal camera devices—smartphones or smart glasses—then submits the captured media to a detection service, completing the imaging–recognition pipeline without any coordination with or awareness of the attacker. The detection service uses SOTA deepfake detection models, which were all trained and calibrated on digital content, and is *not* updated to account for the physical recapture configuration in use. This assumption reflects realistic deployment conditions: a fact-checking API or on-device detection tool has no prior knowledge of which display environment a submitted clip was recorded in, and cannot dynamically recalibrate per submission.

3.2 Representative Scenarios

The display-mediated threat model applies to many ordinary settings where observers do not have access to the original media file and can only verify what is being shown on a screen. The access required to place adversarial content varies across these settings: meetings, classrooms, and conference presentations may be presenter-controlled, whereas public signage and courtroom displays typically require authorization, insider access, or system compromise. We summarize three most outstanding and exemplary concerns below.

Fake News Broadcast in Public Venues. In transportation hubs and shopping malls, a large public screen can show a short video of a deepfake-generated public figure making a statement, and even make the videos mimic a TV news setting. A suspicious bystander cannot access the source file driving the display controller; the only practical option is to record the screen from where they are standing and run a detector on the recaptured clip. The potential consequences of failed detection include immediate public confusion, reputation damage to the impersonated individual, and rapid secondary dissemination when bystanders repost their own recaptured videos to social media as seemingly authentic evidence.

Meeting-room Display Manipulation. In a corporate briefing or conference session, a compromised presentation machine displays a face-swapped video on a meeting-room panel. Audience members who become suspicious may

use their phones to capture a few seconds of the screen for later verification. In critical settings such as corporate merger planning, the impacts of undetected deepfakes range from financial loss to legal consequences once the manipulated recording is circulated as authentic meeting evidence.

Misleading Courtroom Decisions. During a live hearing, a lawyer or third party presents video evidence on a courtroom monitor or evidence display. Although the original file may exist elsewhere in the evidentiary chain, people in the room often encounter the content only through the displayed version and may attempt immediate verification from what is shown on the screen rather than from direct access to the source file. In this case, display-mediated deepfakes affect not only misinformation settings but also formal decision-making environments where provisional trust in displayed evidence can influence high-stakes judicial judgments.

Across these examples, the detection algorithm never receives the original file and must instead verify a camera recapture of displayed content. This breaks the direct-recognition assumption used by existing deepfake detectors, and shows a clear semantic and methodological gap that our work aims to address.

4 Characterization of the Display-Recapture Channel

In display-mediated deepfake attack settings, the detector receives the output of a physical display-recapture channel rather than the generator-produced bitstream itself. To understand the impact of this shift, we begin with a controlled factor characterization that isolates the dominant recapture mechanisms and examines how they alter detector behavior. As illustrated in Figure 2, the recaptured input is shaped by the geometry and photometry of the display-recapture pathway, including factors such as distance, viewing angle, lighting, and display-induced artifacts. The goal of this section is therefore to identify which of these factors most consistently degrade performance, through what mechanisms they do so, and which are sufficiently recurring to justify explicit modeling in the mitigation design developed later in Section 5.

4.1 Experimental Setup

We construct **Dataset D1** from Celeb-DF-v2 [12] by sampling 50 real and 50 fake videos and extracting 5 frames per video for controlled testing. To avoid data leakage, D1 is excluded from the training split used for detector retraining. We study three representative detectors: Xception (spatial CNN) [4], FFD (attention-based CNN) [6], and SPSL (spatial-phase frequency-domain) [13]. All three achieve near-perfect digital-domain performance before recapture, establishing a clean upper bound for degradation analysis. Controlled captures are performed using a fixed display-recapture pair while varying one physical factor at a time. We report both area under curve (AUC) and accuracy (ACC). This distinction is important because recapture may preserve ranking ability while displacing detector scores enough to invalidate a fixed decision threshold.

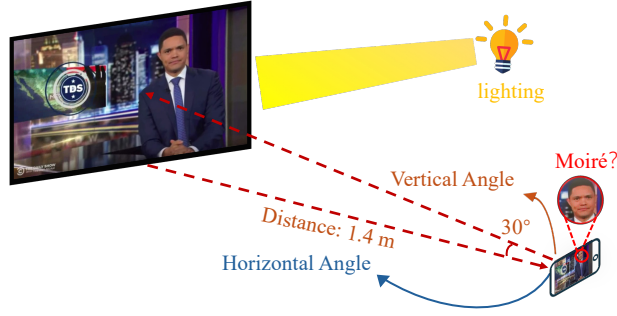


Fig. 2. Typical display-recapture setting considered in the controlled study. Deepfake content is shown on a public display and recaptured by a camera device; the experiments vary factors such as display-recapture distance, viewing angle, lighting, and display-induced artifacts.

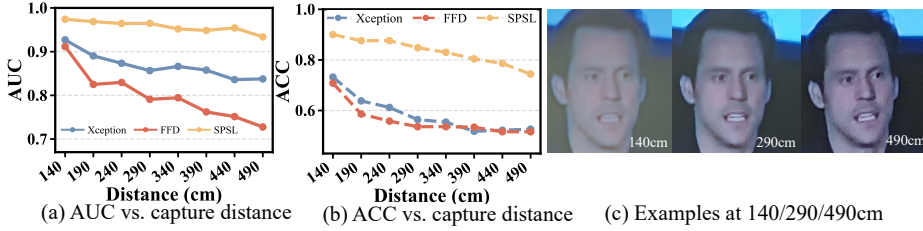


Fig. 3. Effect of distance on detector performance.

The controlled study reveals that detector degradation under recapture is driven primarily by four recurring factors: *distance*, which reduces effective facial resolution; *viewing angle*, which introduces geometric distortion and reflection-dependent changes; *lighting*, which alters photometric statistics through capture quality and camera-side processing; and *Moiré interference*, which injects structured periodic artifacts. The following subsections analyze these four factors individually and show how they affect detector behavior.

4.2 Display-Camera Distance

We evaluate eight display-recapture distances from 140 cm to 490 cm in 50 cm increments under frontal (0°) capture and standard indoor lighting. Figure 3 reports frame-level AUC and ACC as distance increases. All three detectors degrade with distance, but the severity and failure pattern differ.

The dominant mechanism is effective resolution loss. After face detection and alignment, the cropped face must be resized back to the detector’s fixed input resolution (e.g., 299×299 for Xception), which suppresses fine-grained evidence and introduces interpolation artifacts. This especially harms deepfake cues that reside in high-frequency structure, such as blending-boundary inconsistencies, compression irregularities, and synthetic texture artifacts.

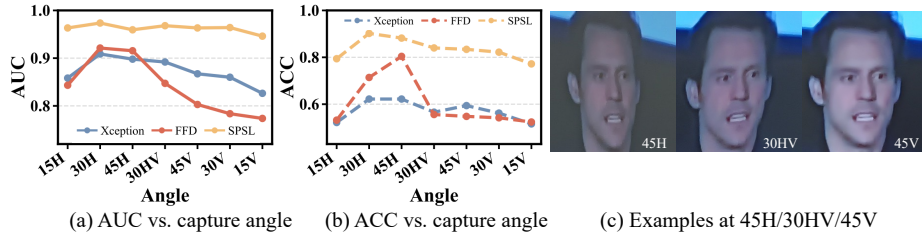


Fig. 4. Effect of viewing angle on detector performance.

SPSL is substantially more robust under the same resolution loss, maintaining ACC = 0.744 even at 490 cm. A plausible explanation is that its spatial-phase representation captures relational frequency structure rather than relying primarily on absolute local texture values. Such relational cues appear better preserved under downsampling than the raw high-frequency evidence on which Xception and FFD depend.

Distance also exposes a clear AUC–ACC dissociation for Xception and FFD once the capture range becomes sufficiently large. Beyond roughly 190 cm, both models retain non-trivial ranking ability while thresholded accuracy deteriorates sharply. For example, at 390 cm, Xception still achieves AUC = 0.858 but falls to ACC = 0.518; FFD reaches AUC = 0.762 with ACC = 0.534. This pattern indicates *calibration collapse*: the detector has not completely lost discrimination, but its score distribution shifts enough that the fixed decision threshold becomes ineffective.

4.3 Viewing Angle

We evaluate seven horizontal and vertical off-axis capture configurations at a fixed display-recapture distance of 240 cm, including 15°H, 30°H, 45°H, 15°V, 30°V, 45°V, and a combined 30°HV condition. Figure 4 reports the corresponding AUC and ACC. Across the three detectors, angle sensitivity is clearly non-uniform and, in several cases, strongly non-monotonic.

The most surprising pattern is that the smallest tested angle, 15°, is often the most damaging. This behavior contradicts the expectation that larger viewing angles should monotonically degrade detector performance through stronger perspective distortion. A more plausible explanation is that the dominant mechanism is not geometric warp alone, but the interaction between viewing geometry and specular reflection. At near-frontal orientations, the camera is more likely to lie close to the reflection direction of the display surface, causing highlights to fall directly on the facial region and shifting detector scores systematically. As the viewing angle increases, these highlights are steered away from the camera. Although perspective distortion becomes more visible, it is spatially smoother and appears less destructive on semantically important facial regions.

We also observe a consistent asymmetry between horizontal and vertical tilt. Vertical tilt is generally more damaging than horizontal tilt of comparable mag-

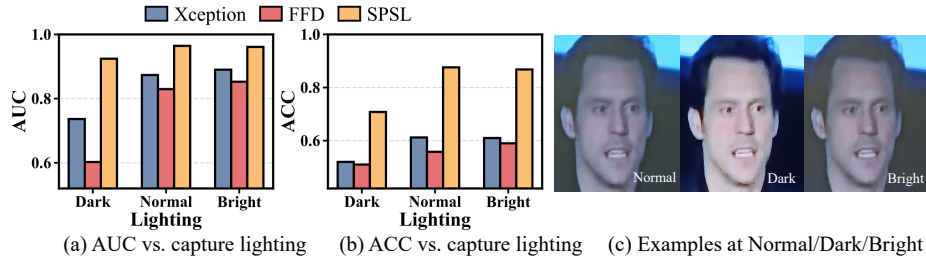


Fig. 5. Effect of lighting condition on detector performance.

nitude. This asymmetry is consistent with face geometry under perspective projection. Vertical tilt compresses the forehead-to-chin extent of the face and alters the vertical proportions of key facial structures, whereas horizontal tilt primarily induces lateral keystoneing and aspect-ratio change. The combined 30° HV condition falls between the corresponding single-axis cases, again indicating that the vertical component is the more disruptive one. Overall, the angle measurements show that recapture degradation depends not only on how much geometric distortion is introduced, but also on how viewing geometry interacts with reflection and detector architecture.

4.4 Lighting

We evaluate three lighting conditions at the standard frontal capture position of 240 cm: *dark* (no ambient light), *normal* (standard indoor lighting), and *bright* (supplemental LED fill light). Figure 5 shows that darkness is the most damaging lighting condition and that the effect is strongly asymmetric. Adding supplemental illumination produces only minor change relative to normal indoor lighting for all three detectors, whereas removing ambient light causes a much larger drop. Under total darkness, Xception falls to $\text{ACC} = 0.520$, and FFD drops to $\text{AUC} = 0.602$, approaching random ranking.

This asymmetry suggests that lighting affects detection less through scene visibility itself than through the way low-light capture alters the camera pipeline. In darker environments, the recaptured image is more likely to undergo noisy acquisition and stronger device-dependent processing, which can smooth or distort the fine-grained spatial evidence used by the detector. This is especially harmful to CNN-based models, which rely heavily on subtle local texture cues.

By contrast, the bright condition yields cleaner recaptures and therefore does not introduce the same level of degradation as the dark condition. SPSL, whose spatial-phase features are more resilient to luminance-driven texture corruption, retains relatively strong ranking ability even under darkness ($\text{AUC} = 0.924$). Its ACC still drops to 0.708, however, indicating that severe low-light recapture can meaningfully disrupt phase-based evidence as well.

4.5 Moiré Patterns

We include one dedicated capture configuration on the display at a distance and viewing angle chosen to induce visible pixel-grid interference. Under this condition, Moiré causes the largest single-factor ACC drop observed in the controlled study: Xception and FFD both fall to $\text{ACC} = 0.512$, while SPSL drops to $\text{ACC} = 0.554$, with corresponding AUCs of 0.706, 0.627, and 0.775, respectively, consistent with prior findings that Moiré can substantially degrade deepfake detector performance under display-recapture conditions [33].

Moiré differs qualitatively from the other factors examined above. Distance, viewing angle, and lighting mainly weaken, blur, or shift existing forensic evidence. Moiré, by contrast, injects strong periodic structure into the recaptured image itself. As a result, all three detectors lose substantial ranking ability, not just thresholded accuracy. This is especially notable for SPSL: unlike in the distance and angle studies, its usual robustness advantage largely disappears, suggesting that structured periodic interference directly corrupts the frequency-domain evidence on which it relies.

However, we also find that Moiré is difficult to reproduce reliably under ordinary passive capture conditions. Across our full data collection in Section 6, only one stable instance was obtained, and doing so required deliberate search over distance and angle to align the display and camera pixel grids. We therefore do not treat Moiré as representative of the typical display-mediated attack path. Instead, we use it as a worst-case diagnostic for structured periodic interference, showing that certain recapture artifacts can damage even the detector family that is otherwise the most robust in our investigation.

4.6 Main Takeaways

The controlled factor analysis unanimously verifies our hypothesis about display-recapture settings’ detrimental impact on existing deepfake detectors, showing the security gap in utilizing existing detection designs for protecting the emerging new threat model. In addition, the dominant distortions identified here are *actionable for mitigation*. Distance, lighting, and viewing angle map naturally onto simulatable training-time operations. The key implication is that a defense need not reproduce every possible recapture environment in full detail; instead, it should model the dominant and repeatable channel mechanisms that account for most of the observed failures. This observation directly motivates the channel-aware design of CoC in the next section.

5 CoC: Channel-Aware Mitigation Design

Guided by the controlled factor characterization, we develop *Caught-on-Camera* (CoC), a mitigation strategy for display-recapture settings. CoC uses selective training-time augmentation to approximate the dominant display-recapture distortions identified in Section 4. It is agnostic to the display-recapture settings and does not require collection of any physical data for retraining.

5.1 Design Principle and Candidate Factor Space

Dominant Distortion Families. The controlled factor characterization in Section 4 shows that recapture degradation is driven by a compact set of recurring channel mechanisms. In particular, the observed failures can be organized into four dominant distortion families: *resolution degradation*, caused primarily by display-recapture distance; *geometric distortion*, induced by off-axis viewing and perspective effects; *photometric variation*, introduced by lighting, exposure adaptation, and device-dependent color processing; and *secondary compression*, introduced when the camera pipeline re-encodes already synthesized content. These four families define the mechanism space that CoC aims to model.

Candidate Augmentative Operations. The full CoC factor table is provided in Appendix A.1 (Table 5). Rather than listing augmentations for their own sake, it maps each retained CoC factor to a specific display-recapture distortion family and its corresponding implementation. Its purpose is to make the CoC design space explicit by showing which channel effects are modeled, how they are realized during training, and how the selected operations jointly cover the dominant recapture mechanisms identified in Section 4.

With this design space in place, the remaining questions are which candidate factors should be retained, how strongly they should be applied, and how they should be composed into a final training policy. We answer these questions through the multi-stage CoC design procedure described next.

5.2 Stage I: Mechanism-Consistent Screening

Stage I asks whether a candidate factor is sufficiently aligned with the target failure mode to justify inclusion in CoC. To answer this, we evaluate each factor independently by training the detector with only that augmentation active at probability $p = 1.0$. Applying a factor at full probability maximally exposes its effect on the learned decision boundary and provides an efficient screening criterion before finer-grained calibration.

All models in this stage are trained from scratch using the split defined in Section 4.1 and evaluated on the large-scale recaptured dataset using frame-level AUC and accuracy. Factors that fail to improve recaptured performance, or that impose unacceptable regression on the digital domain, are discarded from subsequent stages.

5.3 Stage II: Probability Calibration

Passing Stage I does not imply that a factor should be applied aggressively. A transformation that is useful in principle may still become harmful if it is injected too frequently, either because it shifts the training distribution too far from natural digital inputs or because it produces artifacts stronger than those typically introduced by the real display-recapture channel. Stage II therefore calibrates not *whether* a factor is useful, but *how much* of it should be applied during training.

For each factor that survives Stage I, we perform an independent sensitivity sweep over four application probabilities, $p \in \{0.3, 0.5, 0.7, 1.0\}$, with all other factors absent. This isolates the marginal effect of that factor and reveals the point at which additional exposure ceases to improve recaptured robustness.

We then identify the probability that achieves the highest recaptured accuracy and select the *lowest* probability whose performance lies within 0.01 of that peak. This rule preserves nearly all of the recaptured gain while avoiding unnecessary distortion of the clean digital training distribution.

5.4 Stage III: Interaction-Aware Composition

After individual calibration, the remaining challenge is composition. Factors that are useful in isolation are not necessarily beneficial when combined: multiple operations may target overlapping failure modes, or interact in ways that produce unrealistically severe training samples. The goal of Stage III is therefore to identify a *minimal sufficient set* of factors that captures the dominant recapture mechanisms without introducing unnecessary redundancy or distortion.

To do so, we adopt a staged cumulative composition strategy rather than combining all surviving factors by default. We begin with the strongest individually calibrated factor and progressively extend the policy from photometric variation to resolution degradation, geometric distortion, and secondary compression, while retaining the probabilities selected in Stage II and retraining the detector for each cumulative configuration. The intermediate configurations are used to expose interaction effects and are not required to improve fixed-threshold accuracy monotonically.

This non-monotonic criterion accommodates interactions among individually useful transformations. The final policy is selected based on the joint recaptured ACC and AUC while remaining within the allowed regression bound on digital performance. In particular, adding `brightness_contrast` after `pca` changes recaptured Δ AUC only from +0.050 to +0.051, while ACC decreases from 0.856 to 0.744. This AUC–ACC dissociation is consistent with a shift in score calibration under jointly applied photometric augmentations rather than a corresponding loss of ranking ability. The resulting pipeline is the final CoC policy used to retrain the detector for all subsequent large-scale evaluations in Section 6. Together, the three stages determine what to model, how strongly to apply it, and how to compose it into a final training policy. As the next section shows, this design substantially improves detector robustness under real display-recapture conditions without requiring additional physically collected data for retraining.

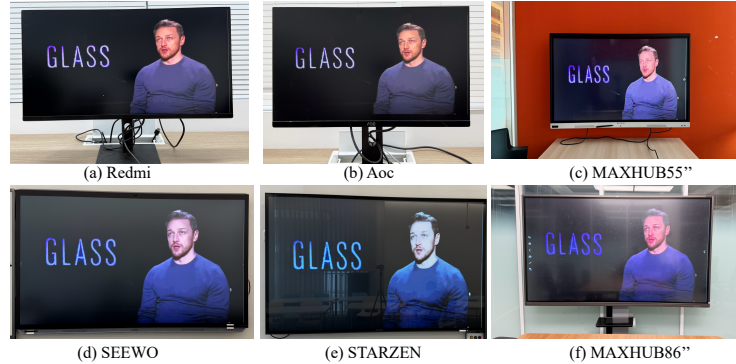
6 Large-Scale Evaluation

6.1 Evaluation Setup

Our evaluation aims to systematically quantify the performance of existing deepfake detectors and CoC’s effectiveness across a wide range of real-world settings.

Table 1. Display and capture hardware used in the large-scale evaluation.

Type	Device	Resolution/Format	Note
Display	Redmi	3840×2160	Workstation monitor
	AOC	1680×1050	Office monitor
	MAXHUB (55")	3840×2160	Study-room
	MAXHUB (86")	3840×2160	Meeting-room
	STARZEN	1920×1080	Classroom
	SEEWO	1920×1080	Discussion-room
Camera	iPhone 17 Pro	4K/30fps	Smartphone
	Vivo V21	4K/30fps	Smartphone
	Meta Wayfarer	1080p/30fps	AI glasses
	Rokid glasses	1080p/30fps	AI glasses

**Fig. 6.** Six display devices.

Dataset. We construct **Dataset D2** by re-sampling (seed = 123) 10 real and 10 fake videos from the Celeb-DF-v2 pool used in Section 4.1, then extracting 5 frames per video (100 frames total). The same three detectors from Section 4.1 are evaluated throughout this section.

Hardware. Table 1 lists the six displays and four capture devices used in the large-scale evaluation, covering desktop monitors, interactive displays, classroom displays, smartphones, and smart glasses.

Configurations and evaluation protocol. We add 84 new recaptured configurations; together with the 19 single-factor settings in Section 4, the paper covers 103 configurations. For each of the $2 \times 6 = 12$ smartphone–display pairs, we collect six settings: frontal (0° , occupancy $\approx 90\%/60\%/30\%$) and oblique (45° , occupancy $\approx 45\%/30\%/15\%$), yielding 72 smartphone datasets. iPhone 17 Pro $\times$ Redmi and Vivo V21 $\times$ Redmi/AOC use D1; other smartphone pairs use D2. Meta Wayfarer and Rokid AI glasses record all six displays under natural usage conditions ($2 \times 6 = 12$, all D1). All frames are face-aligned by dlib; failed detections are removed, with success rate above 95%.

Table 2. Detailed overview of the 103 recaptured configurations. D1 and D2 contribute 500 and 100 frames per configuration, respectively.

Group	Camera–display setting	Capture variation	Set	Cfg.	Frames per cfg.	Total
Distance	Vivo V21 × SEEW0	Eight distances from 140 to 490 cm in 50 cm intervals; frontal capture and normal lighting	D1	8	500	4,000
Lighting	Vivo V21 × SEEW0	Dark, normal, and bright ambient-light conditions; frontal capture at 240 cm	D1	3	500	1,500
Viewing angle	Vivo V21 × SEEW0	15°/30°/45° horizontal, 15°/30°/45° vertical, and combined 30°HV	D1	7	500	3,500
Moiré	Vivo V21 × Redmi	One deliberately induced pixel-grid interference condition	D1	1	500	500
Smartphone						
D1	iPhone 17 Pro × Redmi; Vivo V21 × Redmi/AOC	Six settings per pair: frontal (90/60/30% occupancy) and 45° oblique (45/30/15% occupancy)	D1	18	500	9,000
Smartphone						
D2	Remaining nine smartphone–display pairs	The same six frontal and oblique settings for each pair	D2	54	100	5,400
Smart glasses	Meta Wayfarer and Rokid AI glasses × six displays	Natural-usage captures; one configuration per camera–display pair	D1	12	500	6,000
			Total	103	–	29,900

Table 2 provides a structured overview of the controlled and multi-factor configurations used throughout the evaluation.

6.2 Headline Results of Overall Performance

Across 103 recaptured configurations, digital-domain performance remains high, with D1 frame-level ACC in [0.940, 0.974]. CoC is slightly below the strongest digital baseline, indicating limited clean-domain cost.

Under recapture, unaugmented Xception reaches average AUC and ACC of 0.890 and 0.656, respectively, and FFD is lower at 0.814/0.620. In the hardest STARZEN setting, captured on an iPhone at a 45° viewing angle with approximately 15% screen occupancy, FFD drops to AUC 0.505 and ACC 0.500. SPSL is more robust, with average 0.956/0.836 and worst-case 0.772/0.554, consistent with Section 4. The results verify that CoC substantially improves Xception: average recaptured ACC rises from 0.656 to 0.895 (+0.239), and worst-case performance improves from 0.505/0.500 to 0.775/0.730.

We further evaluate a simple video-level baseline by averaging the five frame-level prediction scores belonging to each recaptured video and applying the same fixed threshold of 0.5. This score-averaging protocol aggregates frame-wise outputs and does not introduce a temporal model. Across the 103 recaptured configurations, unaugmented Xception achieves average video-level AUC and ACC of 0.926 and 0.652, respectively, whereas CoC reaches 0.981 and 0.918. Compared with the corresponding frame-level results, score averaging improves Xception’s

Table 3. Average frame-level ACC per display. CoC denotes Xception trained with the final recapture-aware pipeline.

Method	Redmi	AOC	MAXHUB (55")	MAXHUB (86")	SEEWO	STARZEN
Xception	0.773	0.738	0.678	0.704	0.639	0.516
FFD	0.731	0.714	0.610	0.639	0.576	0.512
SPSL	0.884	0.862	0.876	0.822	0.884	0.718
CoC	0.926	0.912	0.922	0.892	0.917	0.812

ranking ability but does not recover its fixed-threshold accuracy, indicating that calibration collapse persists at the video level. The improvement from CoC therefore remains consistent under video-level aggregation.

6.3 Ablation of CoC Components

Full ablation results are reported in Appendix A.4. In brief, removing any subset of factors from the final CoC pipeline consistently reduces performance, and every tested ablated variant remains below the full model. This indicates that CoC’s gain does not come from a single dominant operation, but from the complementary effect of jointly modeling multiple recapture mechanisms.

6.4 Display-Level Robustness

Table 3 shows that CoC improves ACC on all displays, with larger gains on harder conditions: +0.153 (Redmi), +0.174 (AOC), +0.244 (MAXHUB 55"), +0.188 (MAXHUB 86"), +0.278 (SEEWO), and +0.296 (STARZEN). On the STARZEN display, Xception’s accuracy rises from 0.516 to 0.812. After augmentation, CoC matches or exceeds unaugmented SPSL on four of six displays.

This pattern is consistent with CoC design: `dist` and `geo` are most useful where resolution loss and viewpoint distortion dominate, while photometric factors and `comp_strong` are more beneficial on reflective classroom displays with stronger glare and device-dependent camera processing.

6.5 Configuration-Level Behavior

CoC yields performance gains that are robust to diverse hardware configurations, including those exhibiting non-monotonic recapture behavior. A prominent example is the occupancy inversion where, counter-intuitively, baseline accuracy degrades as the subject moves closer to the display. This is primarily due to larger face crops undergoing more aggressive camera-pipeline processing and compression, which skews detector scores toward the "real" class. For instance, in the iPhone 17 Pro × AOC setup, baseline Xception accuracy drops from 0.91 (30% occupancy) to 0.61 (90%). CoC effectively mitigates this inversion: all configurations maintain accuracy above 0.85, with the 90% occupancy case recovering to 0.92. This resilience stems from `comp_strong` augmentation, which prepares

Table 4. Average frame-level ACC by capture device (all displays and configurations). CoC = Xception trained with the recapture-aware augmentation pipeline.

Method	iPhone 17 Pro	Vivo V21	Meta	Rokid
Xception	0.692	0.662	0.700	0.618
FFD	0.606	0.667	0.611	0.577
SPSL	0.820	0.862	0.872	0.811
CoC	0.896	0.900	0.910	0.874

the model for the intense re-encoding typical of close-range captures. Ultimately, these results demonstrate that CoC prevents calibration collapse across a broad operational envelope rather than just optimizing for a narrow subset of cases.

6.6 Device-Level Generalization

Table 4 shows that the gains from CoC extend across both smartphones and smart glasses. For smartphones, the baseline already indicates device-dependent recapture behavior: FFD differs by 0.061 between iPhone and Vivo (0.606 vs. 0.667), suggesting that the two devices transform captured content in different ways before detection. After augmentation, that gap nearly disappears for CoC, which reaches 0.896 on iPhone and 0.900 on Vivo. This suggests that the photometric components of CoC account for a substantial portion of the Apple–Vivo cross-device variation without requiring device-specific training data.

SPSL is already less sensitive to this cross-phone variation in the baseline, with an iPhone–Vivo gap of 0.042 (0.820 vs. 0.862), smaller than the gap seen in the CNN-based detectors. This is again consistent with the earlier finding that spatial-phase features are less affected by photometric distortion.

For smart glasses, the baseline gap is larger: Xception achieves 0.700 on Meta Wayfarer glasses but only 0.618 on Rokid AI glasses, a difference of 0.082. CoC reduces this gap to 0.036 (0.910 vs. 0.874), but does not eliminate it entirely. This residual difference suggests that the current augmentation recipe does not fully capture Rokid-specific camera-side processing and recapture behavior, and that device-specific recalibration may still be beneficial in practice.

Taken together, the device-level results reinforce the main message of this section. The dominant degradation under recapture is driven more by photometric processing and resolution-related distortion than by any single fixed camera geometry, and a channel-aware training policy can generalize across a heterogeneous set of capture devices, including smart glasses.

6.7 Architectural Implications for Robust Detector Design

The large-scale evaluation suggests that SPSL benefits from its feature space rather than from a favorable data split. Spatial-phase features preserve relational

frequency structure under photometric shift, resolution loss, and moderate geometric distortion. CNN detectors are more tied to absolute texture cues, which are easier to perturb in the display-recapture channel.

This advantage is still conditional. Moire interference directly disturbs frequency structure and can erase much of SPSL’s lead. In severe settings such as STARZEN, SPSL also shows calibration collapse. No current detector is consistently robust across all recapture conditions.

These results support a clear design principle: physical-channel robustness should be optimized explicitly during training. CoC is a practical example. It lifts Xception across displays and devices and often approaches the unaugmented SPSL level. The remaining gap appears to be partly a training-distribution issue, not only an architectural limit.

A pilot multi-stream experiment adds a lightweight FFT branch to an F1-augmented Xception model and trains it jointly. Overall recaptured ACC drops from 0.895 to 0.854, but digital-domain cost is much smaller, with D1 Δ ACC changing from -0.030 to -0.004 , and Moire gain increases from $+0.252$ to $+0.302$. An exploratory **Freq-all** variant reaches 0.890 overall ACC and Moire gain of $+0.350$. Since that setting also changes initialization and schedule, we treat it as suggestive rather than conclusive. Overall, these results support further controlled study of jointly trained spatial–frequency detectors.

7 Discussion & Future Work

This section first synthesizes our findings with respect to the two research questions and then discusses the limitations, remaining challenges, and promising directions for future work.

Regarding RQ1, display recapture is not merely a generic form of image-quality degradation. It affects deepfake detectors through two coupled mechanisms: the erosion of forensic cues and the displacement of detector score distributions. Resolution loss, geometry- and reflection-dependent distortion, photometric and device-side processing shifts, and structured Moiré interference contribute to these mechanisms in different and sometimes non-monotonic ways. Their impact is also architecture-dependent, explaining why useful ranking ability remains even when fixed-threshold accuracy approaches random performance.

Regarding RQ2, effective defenses should explicitly model the display-recapture channel rather than rely on digital-domain generalization. Our results validate selective recapture-aware augmentation as one practical strategy: synthetic channel transformations can improve robustness without using physically recaptured samples as training inputs. More broadly, the coexistence of forensic-cue erosion and score displacement suggests that training-time robustness and deployment-time calibration address complementary aspects of the problem.

7.1 Limitation and Further Defense Exploration

Under the vision of moving toward better security protections against on-screen deepfakes, our paper serves as an exploratory first step that presents a dataset,

threat model, and mitigation design template. At the architectural level, CoC’s training-data augmentation methodology is broadly applicable without modifying detector-specific components, and future work can further validate its empirical effectiveness beyond the Xception-based evaluation presented here. The defense strategies proposed in this work are not intended to be optimal. Instead, they are meant to motivate further systematic innovations in display-recapture-oriented deepfake defense. We highlight several promising directions for future research.

First, the data augmentation algorithms used to approximate individual distortion factors, as well as the method for integrating these augmentation building blocks, can be further improved. The current CoC pipeline models a compact set of dominant factors identified in our controlled study, but each factor is still instantiated using a limited set of standard image transformations. More faithful simulations of specular reflection, device-dependent ISP processing, structured periodic interference, and multi-factor interactions may further improve robustness.

Second, recaptured image correction can be considered as the other side of the coin relative to data augmentation. Compared to the “forward-processing” strategy of augmentation and retraining, this “backward-processing” correction approach may offer the practical benefit of improving detector inputs without modifying existing detector models. However, certain physical recapture distortions are not fully reversible mathematical operations. Therefore, recapture correction alone cannot fully replace the effect of augmentation and retraining. We thus envision a hybrid defense pipeline that combines selective augmentation, recaptured-image correction, and detector-side adaptation.

Third, while the existing CoC pipeline is deliberately agnostic to specific display-recapture pairs, incorporating prior knowledge about the hardware devices should theoretically improve recapture-oriented detection performance. Our results already suggest that different phones and smart glasses introduce different camera-side processing behaviors, and that a residual gap remains even after augmentation. This implies that deployment-specific information, such as display type, camera model, or a small number of reference recaptures, could be used to support device-aware augmentation, threshold recalibration, or lightweight adaptation.

7.2 Challenging Displays: The Case of DLP Projectors

During the course of our evaluations, we found that certain display types, especially Digital Light Processing (DLP) projectors widely used in classrooms and conference rooms, exhibit a substantially different and more challenging display-recapture channel than the flat-panel displays studied in our main experiments. As shown in Figure 7, the camera recaptures often contain strong horizontal color stripes whose shapes and colors can change over time, even when the projected content itself remains unchanged. More importantly, these artifacts are not visible to human observers in the physical scene, but become prominent in the captured images and can significantly alter the detector input.

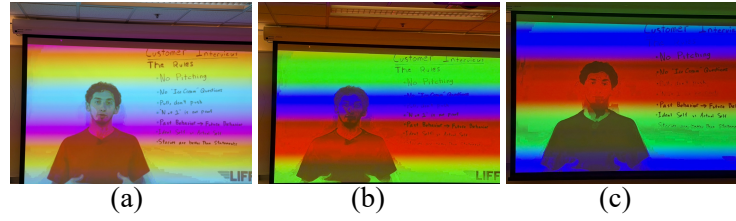


Fig. 7. The same projected content exhibits different horizontal color banding phases at different shooting times with a Panasonic PT-REZ80 Projector, showing an outstanding challenge for projector-camera channels.

We do not yet claim a complete mechanism-level explanation, but a plausible conjecture is the interaction between the rolling shutter of commodity CMOS cameras [10] and the temporally multiplexed color output of DLP projectors. This makes the projector-camera channel qualitatively different from the distortion factors analyzed in Section 4. Although recent camera-sensor security studies have shown that image streams can be shaped by physical-layer interactions with CMOS cameras [46,31,32], to the best of our knowledge, this distinct projector-camera channel effect has not been systematically modeled in prior research, and it presents a unique challenge for both measurement and defense.

This observation also has important security implications. An attacker who understands this display-specific weakness may further exploit projector-based venues to increase the difficulty of user-side verification. For defenders, the main challenge is not only the strength of the chromatic corruption, but also its frame-dependent variability. We therefore believe that effective mitigation will require an explicit model of the projector-camera channel, together with corresponding augmentation or adaptation methods that account for its internal temporal-color behavior, which remains an important direction for future work.

8 Related Work

This section briefly discusses prior works that are the most relevant to our work to provide a holistic view of the problem space.

Display-mediated Attacks and Countermeasures. Our investigation of emerging deepfake threats in public spaces centers on a threat model that utilizes an attacker-controlled screen. Screen displays have previously been identified as a feasible attack vector against various computer vision and deep learning-based algorithms, such as face recognition and authentication [3,25,43,18,16]. However, prior works focused on conventional non-deepfake settings, and showed a gap in analyzing the performance of deepfake detectors in various realistic mobile device camera capture environments. In response to display-mediated presentation attacks, researchers have investigated different ways to identify screen-based attacks [27], such as using light field cameras [26,17] and depth cameras [38]. Other methods use deep neural networks to learn discriminative features of

both display-specific artifacts and liveness cues [15,24]. However, these previous methods require either dedicated sensing hardware or dedicated datasets newly collected for training a discriminative detector model, and are out of the scope of deepfake detections. In contrast, our work investigates a pipeline rooted in the deepfake display setups, that aims to augment existing models without extra training data. Furthermore, our evaluations show how this reusable framework could enable deepfake detection on unseen displays.

Deepfake Image Detection Robustness Analysis. The robustness problem of deepfake detectors has been analyzed in a few previous works that complement our evaluations. For example, Deepfake in The Wild [47] showed the reduced detection accuracies when existing detectors are tested on more diverse datasets collected from the internet. Lu et al. further proposed a framework for systematically measuring how software-space factors, such as image compression and additive noise, could impact deepfake detectors’ performance [19]. Compared to these works that focused on software-space variations, our work presents one of the first systematic investigations on the diverse physical factors and possible countermeasure designs. The most relevant setup to ours is demonstrated by Tariq et al. [33], where the authors focused on the impact of Moiré patterns on deepfake detectors in a similar display-recapture model. Nevertheless, their limited scope on Moiré patterns’ impact, which is encompassed in our research, shows a significant gap in comprehensively analyzing the real-world physical factors. To address this gap, our work also provides the first dedicated analysis of the threat model and possible threat scenarios, as well as actionable security countermeasures that previous works lacked.

Other Modalities of Deepfake Content. While this work aims to provide a pilot analysis on how the emerging display-recapture threat model could strengthen video/image deepfakes, with a focus on face-swapping as a representative example, other types of video deepfake attacks and corresponding defense analysis also exist. For example, GAN and diffusion-based image generation models can create image scenes directly from noise distributions, without modifying any existing images [1,37]. More broadly, there is also increasing offensive and defensive research on audio [42,2,22,40] and textual [28,7,8] deepfake multimedia. While focusing on different modalities of deepfake multimedia from these works, we believe our physical setting modeling methodology and detector augmentation designs could be transferable to other modalities’ settings and benefit future research and designs on physical-world deepfake attacks.

9 Conclusion

This paper presented a systematic study of display-mediated deepfake attacks, showing that the display-recapture channel creates a practical and previously under-studied threat model for user-side deepfake verification. Across 103 recaptured configurations spanning multiple displays, smartphones, and smart glasses, we showed that detectors that perform strongly in the digital domain can degrade severely after physical recapture, in some cases approaching random per-

formance. Our analysis further showed that this degradation is driven by a small set of dominant and recurring factors, including resolution loss, viewing geometry, lighting, and Moiré interference. Guided by these findings, we proposed CoC, a channel-aware mitigation strategy that uses selective augmentation to model the dominant recapture mechanisms without requiring device-specific retraining data. CoC substantially improves robustness, raising Xception’s average frame-level accuracy on recaptured data from 0.656 to 0.895. At the same time, our results show that a meaningful robustness gap remains, especially under high-reflection and projector-based display conditions. We hope this work provides a useful foundation and a wake-up call for future research on physically robust deepfake detection in public communication settings.

Ethical Considerations. This work uses the publicly available Celeb-DF-v2 dataset and recaptured derivatives generated from those benchmark samples for controlled robustness evaluation. No new human-subject photos or videos were collected, and no participants were recruited for this study. The recaptured data were produced by displaying existing benchmark content and recording the displayed output with commodity cameras to emulate realistic verification conditions, rather than by recording new individuals. The goal of this research is purely defensive: to understand the limits of current deepfake detectors and improve their robustness in display-mediated settings. We do not release new personal imagery, and we do not expand the misuse of identity-related content beyond the established benchmark data used for research.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

A Supplementary Materials

All experiments in this appendix use Xception as the detector. Unless otherwise stated, Δ values are computed relative to the no-augmentation baseline. “Rec” denotes the average over 103 recaptured configurations, while D1 and D2 denote performance on the corresponding digital-domain test sets.

A.1 CoC Design Space

Table 5 summarizes the candidate factors considered in CoC, grouped by display-recapture channel family. Each factor is mapped to the channel mechanism it approximates and the training-time operator used to realize it.

A.2 Single-Factor Screening Results

Table 6 reports the single-factor screening results at $p = 1.0$. The six strongest factors are `comp_strong`, `hsv`, `brightness_contrast`, `dist`, `pca`, and `geo`, which are carried forward for probability tuning.

Table 5. CoC design space, grouped by display-recapture channel family. Each candidate factor is instantiated by a training-time operator and then evaluated during CoC selection.

CoC factor	Channel mechanism captured	Training-time realization
RESOLUTION FACTORS		
<code>dist</code>	High-frequency detail loss from display-recapture distance	<code>Downscale</code> (scale: 0.6–0.9)
GEOMETRY FACTORS		
<code>geo</code>	Perspective deformation from off-axis viewing	<code>Perspective</code> (scale: 0.05–0.10)
PHOTOMETRIC FACTORS		
<code>brightness_contrast</code>	Display brightness fluctuation; camera exposure adaptation	<code>RandBrightContrast</code>
<code>hsv</code>	Display color shift; camera white-balance drift	<code>HueSaturationValue</code>
<code>pca</code>	Device-dependent color perturbation during camera capture; sensor response variation	<code>FancyPCA</code>
COMPRESSION FACTORS		
<code>comp_strong</code>	Secondary compression and JPEG-like re-encoding during recapture	<code>ImageCompression</code> (q : 20–70)

Six augmentations produce substantial and consistent gains on recaptured data while incurring only modest or negligible digital-domain cost: `comp_strong`, `hsv`, `brightness_contrast`, `dist`, `pca`, and `geo`. These six factors are retained for probability sensitivity analysis.

A.3 Probability Sensitivity Results

For each retained factor, we sweep the application probability over the distribution $p \in \{0.3, 0.5, 0.7, 1.0\}$ while holding all other factors absent. Table 7 reports the full sweep.

Two distinct sensitivity patterns emerge. `comp_strong` improves monotonically with p , reaching its highest recaptured performance at $p = 1.0$, but also incurring the largest digital-domain cost of any factor tested. By contrast, the other factors peak at moderate or low probability, indicating that overly frequent application can push the training distribution away from realistic recapture conditions. The final selected probabilities are therefore `pca` ($p = 0.5$), `brightness_contrast` ($p = 0.5$), `hsv` ($p = 0.3$), `dist` ($p = 0.3$), `geo` ($p = 0.3$), and `comp_strong` ($p = 0.7$).

Table 6. Single-factor screening results (Xception, frame-level Δ relative to the no-augmentation baseline). Rec: average over 103 recaptured configurations; D1/D2: digital test sets. Factors are sorted by Rec Δ ACC.

Augmentation	Rec Δ ACC	Rec Δ AUC	D1 Δ AUC	D2 Δ ACC
comp_strong	+0.188	+0.052	−0.007	−0.060
hsv	+0.167	+0.058	−0.001	±0
brightness_contrast	+0.137	+0.051	−0.002	±0
dist	+0.113	+0.038	−0.001	−0.010
pca	+0.106	+0.054	−0.001	±0
geo	+0.079	+0.032	−0.003	−0.010

A.4 Final Pipeline and Ablation

Table 8 reports the staged cumulative composition path used to construct the final CoC pipeline.

The final CoC pipeline is:

- `pca` ($p = 0.5$), `brightness_contrast` ($p = 0.5$), and `hsv` ($p = 0.3$),
- `dist` ($p = 0.3$) and `geo` ($p = 0.3$),
- `comp_strong` ($p = 0.7$).

This configuration achieves abs ACC = 0.895 (+0.239 over baseline), Rec Δ AUC = +0.075, improves all 103 recaptured datasets, and eliminates calibration collapse on the benchmark.

Table 9 reports ablation results for the final pipeline. All ablated variants fall below the full model, confirming that the gain comes from the complementary effect of jointly modeling multiple recapture mechanisms.

The candidate operators are all standard image augmentations. Accordingly, the single-factor experiments in Tables 6 and 7 serve as standard single-operation augmentation baselines, while Table 8 evaluates simpler cumulative compositions. The best individual configuration reaches an ACC of 0.856, and the best incomplete cumulative composition reaches 0.880, both below the final CoC policy at 0.895. These results indicate that the benefit of CoC derives from the calibrated composition of complementary recapture mechanisms rather than from any individual transformation.

Table 7. Probability sensitivity sweep (Xception, frame-level). Bold rows indicate the selected probability for each factor. Δ values are relative to the no-augmentation baseline.

Factor	p	abs ACC Rec	Δ ACC Rec	Δ AUC D1	Δ ACC D2	Δ ACC
comp_strong	0.3	0.730	+0.074	-0.029	-0.006	-0.010
	0.5	0.713	+0.057	+0.038	± 0	-0.010
	0.7	0.746	+0.090	+0.047	± 0	-0.020
	1.0	0.844	+0.188	+0.052	-0.046	-0.060
pca	0.3	0.774	+0.118	+0.059	± 0	-0.010
	0.5	0.856	+0.200	+0.050	-0.004	-0.030
	0.7	0.801	+0.145	+0.026	+0.008	± 0
	1.0	0.762	+0.106	+0.054	+0.006	± 0
brightness _contrast	0.3	0.705	+0.049	+0.020	-0.006	-0.010
	0.5	0.831	+0.175	+0.048	-0.004	-0.010
	0.7	0.801	+0.145	+0.042	-0.004	-0.020
	1.0	0.793	+0.137	+0.051	+0.004	± 0
hsv	0.3	0.829	+0.173	+0.047	+0.004	-0.010
	0.5	0.811	+0.155	+0.049	+0.002	-0.010
	0.7	0.686	+0.030	+0.033	+0.004	-0.010
	1.0	0.823	+0.167	+0.058	+0.004	± 0
dist	0.3	0.802	+0.146	+0.032	+0.002	-0.010
	0.5	0.734	+0.078	+0.039	+0.008	-0.010
	0.7	0.620	-0.036	-0.005	+0.004	-0.010
	1.0	0.769	+0.113	+0.038	+0.012	-0.010
geo	0.3	0.742	+0.086	+0.028	-0.002	± 0
	0.5	0.713	+0.057	+0.035	+0.004	± 0
	0.7	0.738	+0.082	+0.013	+0.010	-0.010
	1.0	0.735	+0.079	+0.032	-0.016	-0.010

Table 8. Staged cumulative composition path (Xception, frame-level). Δ values are relative to the no-augmentation baseline.

Configuration	abs ACCRec	Δ ACCRec	Δ AUCD1	Δ ACCD2	Δ ACC
pca (0.5)	0.856	+0.200	+0.050	-0.004	-0.030
+brightness (0.5)	0.744	+0.088	+0.051	+0.002	-0.010
+hsv (0.3)	0.849	+0.193	+0.064	+0.004	-0.010
+dist (0.3)	0.776	+0.120	+0.054	± 0	± 0
+geo (0.3)	0.880	+0.224	+0.070	-0.006	-0.010
+comp_strong (0.7)	0.895	+0.239	+0.075	-0.030	-0.040
+comp_strong (1.0)	0.871	+0.215	+0.068	-0.022	± 0

Table 9. Ablation test: variants of the final CoC pipeline with individual factors removed (Xception, frame-level).

Variant Configuration		abs ACC Δ vs F1	
CoC	pca + bc + hsv + dist + geo + comp (0.7)	0.895	—
X	pca + dist + geo + comp (0.7)	0.820	−0.075
Z	pca + hsv + geo + comp (0.7)	0.808	−0.087
Y	pca + hsv + dist + geo + comp (0.7)	0.799	−0.096

References

1. Abdullah, S.M., Cheruvu, A., Kanchi, S., Chung, T., Gao, P., Jadliwala, M., Viswanath, B.: An analysis of recent advances in deepfake image detection in an evolving threat landscape. In: 2024 IEEE Symposium on Security and Privacy (SP). pp. 91–109. IEEE (2024)
2. Almutairi, Z., Elgibreen, H.: A review of modern audio deepfake detection methods: challenges and future directions. *Algorithms* **15**(5), 155 (2022)
3. Bhattacharjee, S., Mohammadi, A., Anjos, A., Marcel, S.: Recent advances in face presentation attack detection. *Handbook of biometric anti-spoofing: presentation attack detection* pp. 207–228 (2019)
4. Chollet, F.: Xception: Deep learning with depthwise separable convolutions. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 1251–1258 (2017). <https://doi.org/10.1109/CVPR.2017.195>
5. Cole, S.: Reddit just shut down the deepfakes subreddit. <https://www.vice.com/en/article/reddit-shuts-down-deepfakes/> (Feb 2018), vICE. Accessed: 2026-04-17
6. Dang, H., Liu, F., Stehouwer, J., Liu, X., Jain, A.K.: On the detection of digital face manipulation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 5781–5790 (2020). <https://doi.org/10.1109/CVPR42600.2020.00582>
7. Fagni, T., Falchi, F., Gambini, M., Martella, A., Tesconi, M.: Tweepfake: About detecting deepfake tweets. *Plos one* **16**(5), e0251415 (2021)
8. Go, W., Kim, H., Oh, A., Kim, Y.: Xdac: Xai-driven detection and attribution of llm-generated news comments in korean. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 22728–22750 (2025)
9. Goodfellow, I.J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial nets. *Advances in neural information processing systems* **27** (2014)
10. Köhler, S., Lovisotto, G., Birnbach, S., Baker, R., Martinovic, I.: They see me rollin’: Inherent vulnerability of the rolling shutter in cmos image sensors. In: *Proceedings of the 37th Annual Computer Security Applications Conference*. pp. 399–413 (2021)
11. Li, Y., Wang, H., Shao, Y., Zhang, S., Long, Y.: Devil in the lens: Analyzing and defending physical prompt injection against vision-language models on wearable devices. *arXiv preprint arXiv:2607.10269* (2026)
12. Li, Y., Yang, X., Sun, P., Qi, H., Lyu, S.: Celeb-df: A large-scale challenging dataset for deepfake forensics. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 3207–3216 (2020). <https://doi.org/10.1109/CVPR42600.2020.00327>
13. Liu, H., Li, X., Zhou, W., Chen, Y., He, Y., Xue, H., Zhang, W., Yu, N.: Spatial-phase shallow learning: Rethinking face forgery detection in frequency domain. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 772–781 (2021). <https://doi.org/10.1109/CVPR46437.2021.00084>
14. Liu, K., Perov, I., Gao, D., Chervoniy, N., Zhou, W., Zhang, W.: Deepfacelab: Integrated, flexible and extensible face-swapping framework. *Pattern Recognition* **141**, 109628 (2023)

15. Liu, Y., Jourabloo, A., Liu, X.: Learning deep models for face anti-spoofing: Binary or auxiliary supervision. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 389–398 (2018)
16. Long, Y., Fu, K.: Side auth: Synthesizing virtual sensors for authentication. arXiv preprint arXiv:2301.11745 (2023)
17. Long, Y., Naghavi, P., Kojusner, B., Butler, K., Rampazzi, S., Fu, K.: Side eye: Characterizing the limits of pov acoustic eavesdropping from smartphone cameras with rolling shutters and movable lenses. In: 2023 IEEE symposium on security and privacy (SP). pp. 1857–1874. IEEE (2023)
18. Long, Y., Yan, C., Xiao, S., Prasad, S., Xu, W., Fu, K.: Private eye: On the limits of textual screen peeking via eyeglass reflections in video conferencing. In: 2023 IEEE Symposium on Security and Privacy (SP). pp. 3432–3449. IEEE (2023)
19. Lu, Y., Ebrahimi, T.: Assessment framework for deepfake detection in real-world situations. EURASIP Journal on Image and Video Processing **2024**(1), 6 (2024)
20. Meng, X., Wang, L., Guo, S., Ju, L., Zhao, Q.: Ava: Inconspicuous attribute variation-based adversarial attack bypassing deepfake detection. In: 2024 IEEE Symposium on Security and Privacy (SP). pp. 74–90. IEEE (2024)
21. Mirsky, Y., Lee, W.: The creation and detection of deepfakes. ACM Computing Surveys **54**(1) (Jul 2021). <https://doi.org/10.1145/3425780>, publisher Copyright: © 2021 ACM.
22. Mukhopadhyay, D., Shirvanian, M., Saxena, N.: All your voices are belong to us: Stealing voices to fool humans and machines. In: European Symposium on Research in Computer Security. pp. 599–621. Springer (2015)
23. NBC News: Deepfake videos of tom cruise went viral. their creator hopes they boost awareness. <https://www.nbcnews.com/tech/tech-news/creator-viral-tom-cruise-deepfakes-speaks-rcna356> (Mar 2021), accessed: 2026-04-17
24. Patel, K., Han, H., Jain, A.K., Ott, G.: Live face video vs. spoof face video: Use of moiré patterns to detect replay video attacks. In: 2015 International Conference on Biometrics (ICB). pp. 98–105. IEEE (2015)
25. Phan, B., Mannan, F., Heide, F.: Adversarial imaging pipelines. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 16051–16061 (2021)
26. Raghavendra, R., Raja, K.B., Busch, C.: Presentation attack detection for face recognition using light field camera. IEEE Transactions on Image Processing **24**(3), 1060–1075 (2015)
27. Ramachandra, R., Busch, C.: Presentation attack detection methods for face recognition systems: A comprehensive survey. ACM Computing Surveys (CSUR) **50**(1), 1–37 (2017)
28. Rosca, C.M., Stancu, A., Iovanovici, E.M.: The new paradigm of deepfake detection at the text level. Applied Sciences **15**(5), 2560 (2025)
29. Rößler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., Nießner, M.: Face-forensics++: Learning to detect manipulated facial images. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 1–11 (2019). <https://doi.org/10.1109/ICCV.2019.00009>
30. Sandoval, M.P., de Almeida Vau, M., Solaas, J., Rodrigues, L.: Threat of deepfakes to the criminal justice system: a systematic review. Crime Science **13**(1), 41 (2024)
31. Shamsi, N., Chandrasekar, K., Long, Y., Limbach, C., Rebello, K., Fu, K.: Wip: threat modeling laser-induced acoustic interference in computer vision-assisted vehicles. In: Symposium on Vehicles Security and Privacy (VehicleSec) (2024)

32. Shamsi, N., Long, Y., Fu, K.: Eyehearyou: Probing location identification via occluded smartphone cameras and ultrasound. In: 2025 IEEE International Symposium on Hardware Oriented Security and Trust (HOST). pp. 23–34. IEEE (2025)
33. Tariq, R., Heo, M., Woo, S.S., Tariq, S.: Beyond the screen: Evaluating deepfake detectors under moire pattern effects. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 4429–4439 (2024)
34. Tolosana, R., Vera-Rodriguez, R., Fierrez, J., Morales, A., Ortega-Garcia, J.: Deepfakes and beyond: A survey of face manipulation and fake detection. *Information fusion* **64**, 131–148 (2020)
35. Vaccari, C., Chadwick, A.: Deepfakes and disinformation: Exploring the impact of synthetic political video on deception, uncertainty, and trust in news. *Social media+ society* **6**(1), 2056305120903408 (2020)
36. Wakefield, J.: Deepfake presidents used in russia-ukraine war. <https://www.bbc.com/news/technology-60780142> (Mar 2022), BBC News. Accessed: 2026-04-17
37. Wang, X., Huang, J., Ma, S., Nepal, S., Xu, C.: Deepfake disrupter: The detector of deepfake is my friend. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 14920–14929 (2022)
38. Wang, Y., Nian, F., Li, T., Meng, Z., Wang, K.: Robust face anti-spoofing with depth information. *Journal of Visual Communication and Image Representation* **49**, 332–337 (2017)
39. Wei, M., Yeung, C., Roesner, F., Kohno, T.: " we're utterly ill-prepared to deal with something like this": Teachers' perspectives on student generation of synthetic nonconsensual explicit imagery. In: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. pp. 1–18 (2025)
40. Yan, C., Long, Y., Ji, X., Xu, W.: The catcher in the field: A fieldprint based spoofing detection for text-independent speaker verification. In: Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security. pp. 1215–1229 (2019)
41. Yan, Z., Zhang, Y., Yuan, X., Lyu, S., Wu, B.: Deepfakebench: A comprehensive benchmark of deepfake detection. In: Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track (2023), <https://openreview.net/forum?id=hizSx8pf0U>
42. Yi, J., Wang, C., Tao, J., Zhang, X., Zhang, C.Y., Zhao, Y.: Audio deepfake detection: A survey. arXiv preprint arXiv:2308.14970 (2023)
43. Yuan, S., Li, H., Zhang, R., Cao, H., Jiang, W., Ni, T., Fan, W., Zhao, Q., Xu, G.: Omni-angle assault: An invisible and powerful physical adversarial attack on face recognition. In: Forty-second International Conference on Machine Learning (2025)
44. Zhang, T.: Deepfake generation and detection, a survey. *Multimedia Tools and Applications* **81**(5), 6259–6276 (2022)
45. Zhang, X., Tu, Y., Long, Y., Shan, L., Elsaadani, M.A., Fu, K., Lin, Z., Hei, X.: From virtual touch to tesla command: Unlocking unauthenticated control chains from smart glasses for vehicle takeover. In: 2024 IEEE Symposium on Security and Privacy (SP). pp. 2366–2384. IEEE (2024)
46. Zhuang, H., Long, Y., Fu, K.: Rf-eye-d: Probing feasibility of cmos camera watermarking with radio-frequency injection. In: 2025 28th International Symposium on Research in Attacks, Intrusions and Defenses (RAID). pp. 850–865. IEEE (2025)
47. Zi, B., Chang, M., Chen, J., Ma, X., Jiang, Y.G.: Wilddeepfake: A challenging real-world dataset for deepfake detection. In: Proceedings of the 28th ACM international conference on multimedia. pp. 2382–2390 (2020)